

Roll No.....

Mid Semester Examination (Even Semester)
March 2025

Name of the Program: B.Tech.
Name of the Course: Engineering Chemistry

Semester: 2
Course Code: TCH201

Time: 90 Minutes

Maximum Marks: 50

Note:

- i Answer **all the questions** by choosing **any one of the sub questions**.
- ii Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Using molecular orbital theory, compare the bond energy and magnetic character of CO, CO ⁺ and CO ⁻ species.	
OR		
(b)	Write note on nanosized materials.	
Q2	(10 marks)	CO1
(a)	(i) Differentiate between bonding and antibonding molecular orbitals. (ii) Differentiate between intermolecular and intra molecular hydrogen bonding	
OR		
(b)	What do you mean by functional group and fingerprint regions. Explain different types of vibrations involved in IR Spectroscopy.	
Q3	(10 marks)	CO2
(a)	Discuss ion exchange process for water softening with its benefits, and limitations.	
OR		
(b)	Calculate the amount of lime (80% pure) and soda (90% pure) required for treatment of 20,000 litres of water containing: Mg (HCO ₃) ₂ = 9.1, CaCl ₂ = 13.8, Ca (HCO ₃) ₂ = 10.2, MgCl ₂ =11.0, MgSO ₄ = 14.0 CaSO ₄ = 18.0 and NaCl=25	
Q4	(10 marks)	CO2
(a)	Write note on (i) Reverse osmosis (ii) Caustic embrittlement	
OR		
(b)	(i) A completely exhausted zeolite softener requires 40 liters of NaCl solution having 20 g/l of NaCl. Calculate the hardness of water if 15000 liters of water was softened by the zeolite softener. (i) Discuss Boiler problems due to hard water.	
Q5	(10 marks)	CO2
(a)	Explain lime soda process used for softening of hard water by giving suitable reactions. Also write difference between hot and cold lime soda process.	
OR		
(b)	Write postulates of molecular orbital theory and explain linear combination of atomic orbitals.	CO1

Washington, D.C. 20540
U.S. Department of State

Mr. [Name] [Address]
[City] [State] [Zip]

Dear Mr. [Name]:

I am pleased to inform you that your application for a passport has been approved. The passport will be ready for pickup at the U.S. Department of State, Office of Passport Services, on [Date].

Please bring your current identification and the fee of \$[Amount] with you. If you are unable to appear in person, you may authorize a representative by completing Form DS-3047.

For more information, please contact the U.S. Department of State, Office of Passport Services, at [Phone Number].

Sincerely,
[Signature]

Enclosed for you are two copies of the passport application form and a copy of the fee schedule.

Very truly yours,
[Signature]